

xeCJK command-boundary oracle: kernel \ref

Direct input is the oracle. The candidate must have the same visible content and the same boundary spacing. For visibility this MWE uses CJKecglue=5pt and CJKglue=1pt; the red value is candidate – oracle.

| Candidate source                        | Direct-input oracle | Command candidate | Width delta |
|-----------------------------------------|---------------------|-------------------|-------------|
| Western/numeric output between CJK text |                     |                   |             |
| 中\ref {matrix-western} 文                | <div>中 1 文</div>    | <div>中 1文</div>   | -5.0pt      |
| 中 \ref {matrix-western} 文               | <div>中 1 文</div>    | <div>中 1文</div>   | -5.0pt      |
| 中\ref {matrix-western} 文                | <div>中 1 文</div>    | <div>中 1 文</div>  | same        |
| 中 \ref {matrix-western} 文               | <div>中 1 文</div>    | <div>中 1 文</div>  | same        |
| CJK output between Western text         |                     |                   |             |
| A\ref {matrix-chinese}B                 | <div>A 中文 B</div>   | <div>A 中文B</div>  | -5.0pt      |
| A \ref {matrix-chinese}B                | <div>A 中文 B</div>   | <div>A 中文B</div>  | -5.0pt      |
| A\ref {matrix-chinese} B                | <div>A 中文 B</div>   | <div>A 中文 B</div> | -1.35365pt  |
| A \ref {matrix-chinese} B               | <div>A 中文 B</div>   | <div>A 中文 B</div> | -1.35365pt  |
| CJK output between CJK text             |                     |                   |             |
| 中\ref {matrix-chinese} 文                | <div>中中文文</div>     | <div>中中文文</div>   | -1.0pt      |
| 中 \ref {matrix-chinese} 文               | <div>中中文文</div>     | <div>中中文文</div>   | same        |

Known cause: LaTeX’s \setref appends \null after the reference text. The resulting 0pt × 0pt hbox can hide xeCJK’s final boundary marker; see CTeX-org/ctex-kit#991 and #992.